LAB 05 / CLOUD INFRASTRUCTURE

AWS Storage, Databases, and High Availability

EBS, Amazon RDS, load balancing, and Auto Scaling

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Executive Summary

Documented AWS exercises covering persistent block storage, a managed database, and a load-balanced application architecture. The work includes storage and snapshot procedures, database connectivity, launch templates, health checks, and scaling under load.

> **SKILLS DEMONSTRATED**
>
> Amazon EBS · Amazon RDS · Application Load Balancer · EC2 Auto Scaling · CloudWatch

- Documented EBS volume and snapshot workflows.
- Configured an RDS database and connected it to the lab application.
- Recorded load-balancer health checks and CPU-based Auto Scaling behavior.

Evidence Included

EBS procedures, RDS configuration and application screenshots, load-balancer target health, and Auto Scaling observations.

Objective

The objective was to work with persistent storage, deploy a managed database, and configure an application architecture that distributes traffic and scales with demand.

Technical Background

Amazon Elastic Block Store (EBS) provides persistent block storage for EC2 instances. Volumes can retain data independently of an instance stop or restart, while snapshots support backup and restoration. The lab covers volume management and snapshot workflows.

A database server stores and serves structured application data. Amazon Relational Database Service (RDS) manages infrastructure tasks for supported relational database engines. This exercise configures an RDS database and connects a lab web application to it.

Load balancing distributes application traffic across multiple targets. Auto Scaling adjusts instance capacity in response to demand. The exercise combines an Application Load Balancer, launch template, Auto Scaling group, and CloudWatch metrics to examine availability and scaling behavior.

Procedure and Evidence

Lab 4: Working with EBS

Task 1: Create an EBS Volume

Open EC2 from the AWS Management Console.

Select Instances and locate the Lab instance.

Note the instance's Availability Zone (for example, us-east-1a).

In the left navigation pane, choose Volumes.

Select Create volume and configure:

Volume type: General Purpose SSD (gp2)

Size: 1 GiB

Availability Zone: Same AZ as the Lab instance

Add a tag:

Key: Name

Value: My Volume

Choose Create volume.

Wait for the volume state to change from Creating to Available.

Task 2: Attach the Volume to the EC2 Instance

Select My Volume.

Choose Actions → Attach volume.

Configure:

Instance: Lab

Device name: /dev/sdb

Choose Attach volume.

Verify that the volume state changes to In-use.

Task 3: Connect to the EC2 Instance

Return to Instances and select the Lab instance.

Choose Connect.

Open the Session Manager tab and select Connect.

In the terminal, switch to the ec2-user account:

sudo su -l ec2-user

Task 4: Create and Configure a File System

Check Existing Storage

Display current storage devices:

df -h

Only the original 8 GiB root volume should appear.

Format the New Volume

Create an ext3 file system on the new volume:

sudo mkfs -t ext3 /dev/sdb

Create a Mount Point

Create a directory for the new volume:

sudo mkdir /mnt/data-store

Mount the Volume

Mount the volume:

sudo mount /dev/sdb /mnt/data-store

Configure Automatic Mounting

Add the volume to /etc/fstab:

echo "/dev/sdb /mnt/data-store ext3 defaults,noatime 1 2" | sudo tee -a /etc/fstab

Verify the configuration:

cat /etc/fstab

Verify Storage

Check storage again:

df -h

You should now see /dev/xvdb mounted at /mnt/data-store.

Create and Verify a Test File

Create a text file on the volume:

sudo sh -c "echo some text has been written > /mnt/data-store/file.txt"

Confirm the file contents:

cat /mnt/data-store/file.txt

Task 5: Create an EBS Snapshot

In the EC2 Console, select Volumes and choose My Volume.

Select Actions → Create snapshot.

Add a tag:

Key: Name

Value: My Snapshot

Choose Create snapshot.

Open Snapshots from the navigation pane.

Wait for the snapshot status to change from Pending to Completed.

Remove the Test File

In the terminal, delete the file:

sudo rm /mnt/data-store/file.txt

Verify the file has been removed:

ls /mnt/data-store/

Task 6: Restore the Snapshot

Create a Volume from the Snapshot

Select My Snapshot.

Choose Actions → Create volume from snapshot.

Configure:

Availability Zone: Same AZ used previously

Add a tag:

Key: Name

Value: Restored Volume

Choose Create volume.

Attach the Restored Volume

Open Volumes and select Restored Volume.

Choose Actions → Attach volume.

Configure:

Instance: Lab

Device name: /dev/sdc

Choose Attach volume.

Verify the volume state changes to In-use.

Mount the Restored Volume

Create a mount directory:

sudo mkdir /mnt/data-store2

Mount the restored volume:

sudo mount /dev/sdc /mnt/data-store2

Verify the Restored Data

List the contents of the restored volume:

ls /mnt/data-store2/

Confirm that file.txt is present.

This demonstrates that the EBS snapshot successfully preserved the data and restored it to a new volume.

Lab 5: Build a Database Server

Task 1: Create a Security Group for the Database

Open VPC from the AWS Management Console.

In the left navigation pane, select Security Groups.

Choose Create security group and configure:

Security group name: DB Security Group

Description: Permit access from Web Security Group

VPC: Lab VPC

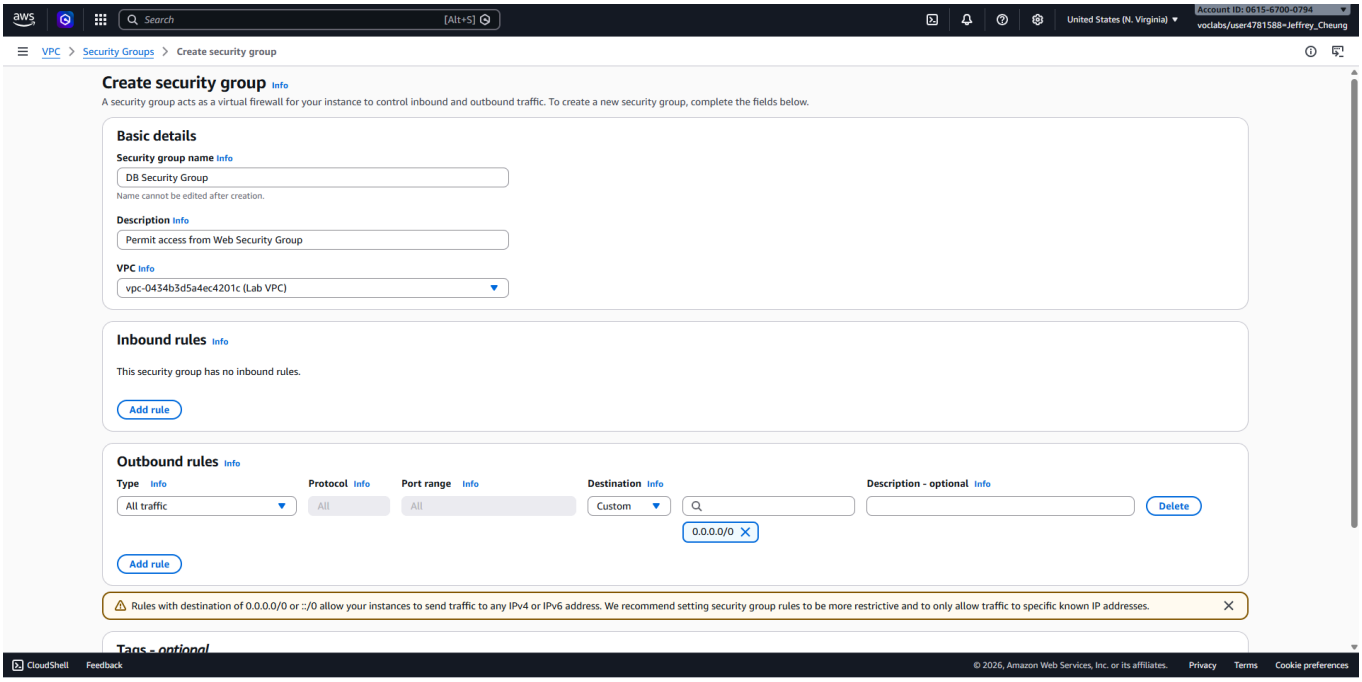


Figure 01. VPC: Lab VPC

Under Inbound rules, choose Add rule.

Configure the rule:

Type: MySQL/Aurora (3306)

Source: Web Security Group

Choose Create security group.

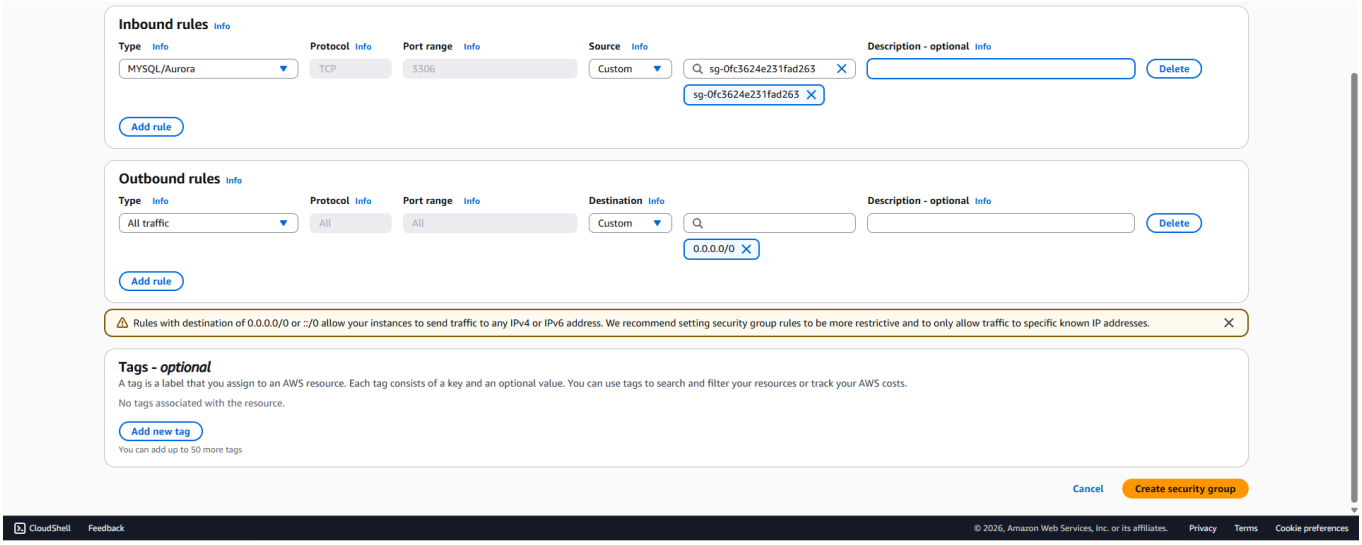


Figure 02. Choose Create security group.

This security group allows EC2 instances associated with the Web Security Group to connect to the database on port 3306.

Task 2: Create a DB Subnet Group

Open RDS from the AWS Management Console.

In the left navigation pane, select Subnet groups.

Choose Create DB subnet group.

Configure:

Name: DB-Subnet-Group

Description: DB Subnet Group

VPC: Lab VPC

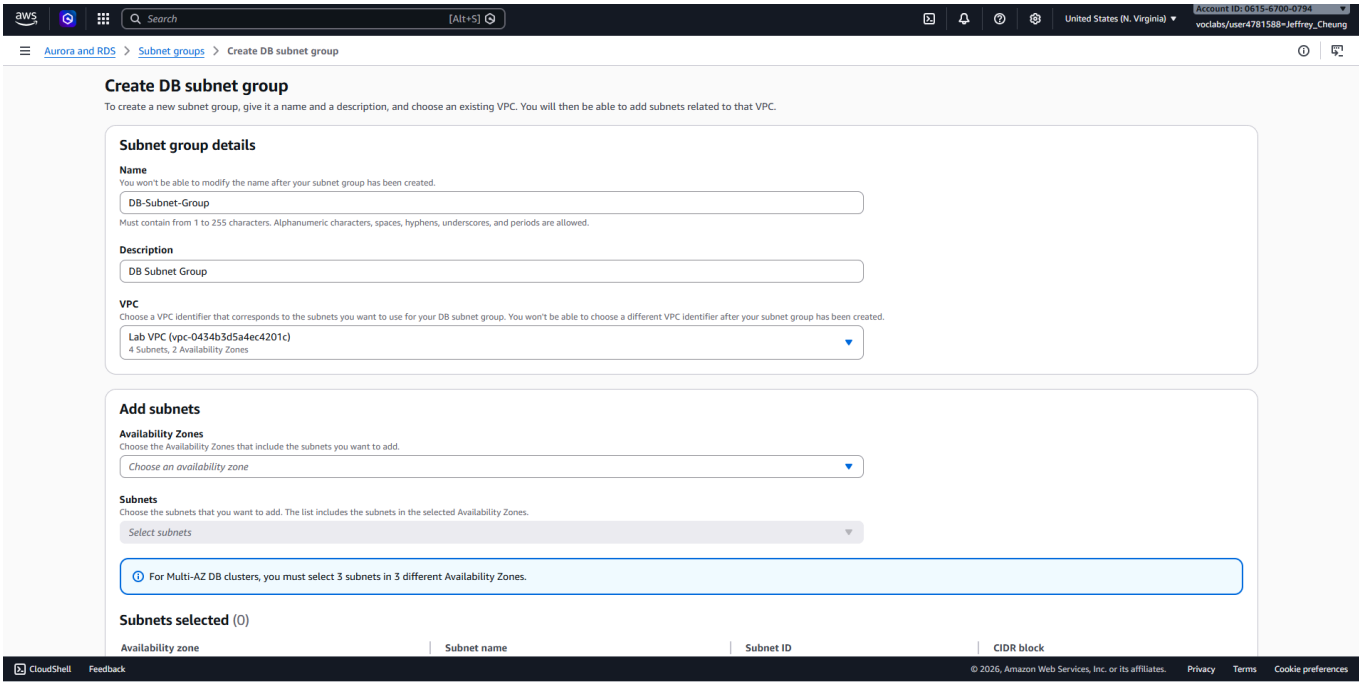


Figure 03. VPC: Lab VPC

Under Availability Zones, select:

us-east-1a

us-east-1b

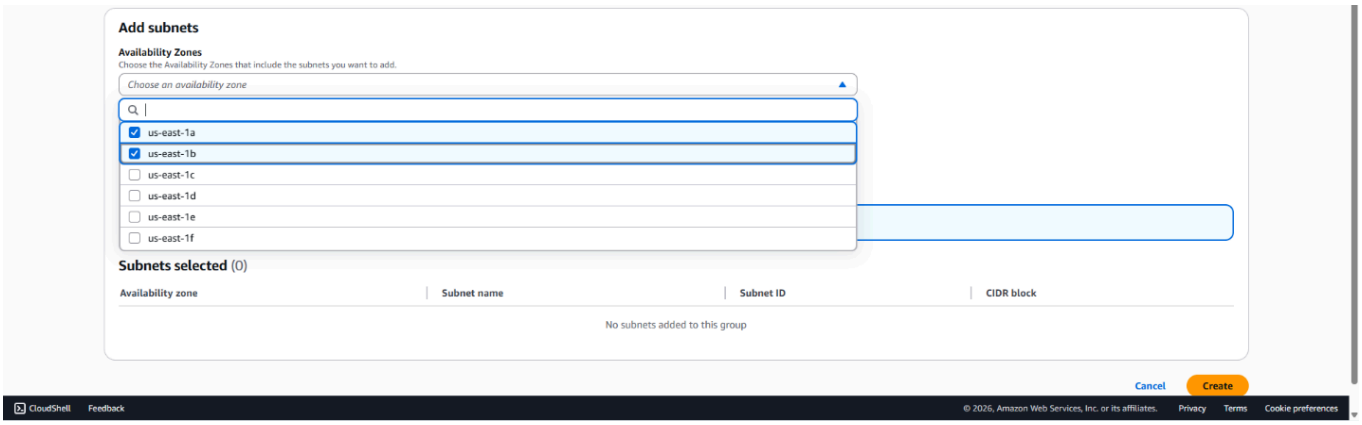


Figure 04. us-east-1b

Under Subnets, select the private subnets with CIDR ranges:

10.0.1.0/24

10.0.3.0/24

Verify that both subnets appear in the selected subnet list.

Choose Create.

The subnet group defines which subnets Amazon RDS can use when deploying the database.

Task 3: Create an Amazon RDS Database

In the left navigation pane, select Databases.

Choose Create database.

Under Engine options, select:

MySQL

Under Templates, choose:

Dev/Test

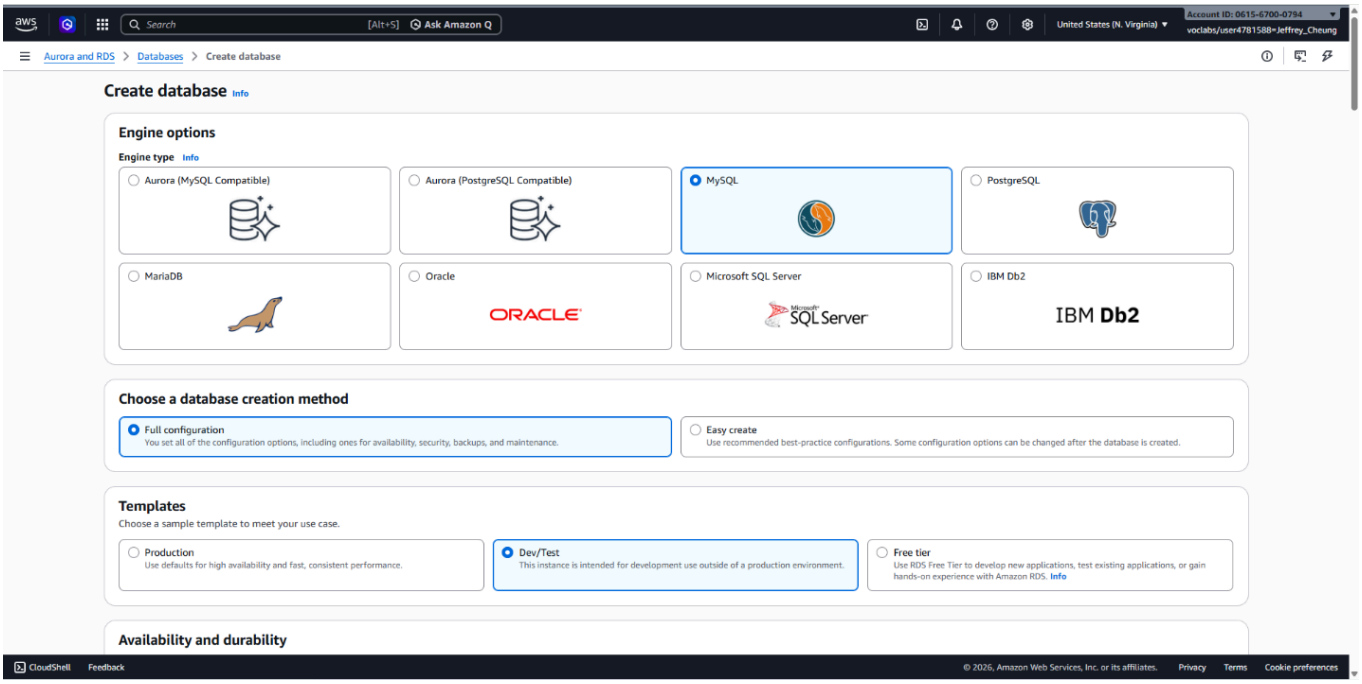


Figure 05. Dev/Test

Under Availability and durability, select:

Multi-AZ DB instance

Database Settings

Configure:

DB instance identifier: lab-db

Master username: main

Master password: lab-password

Confirm password: lab-password

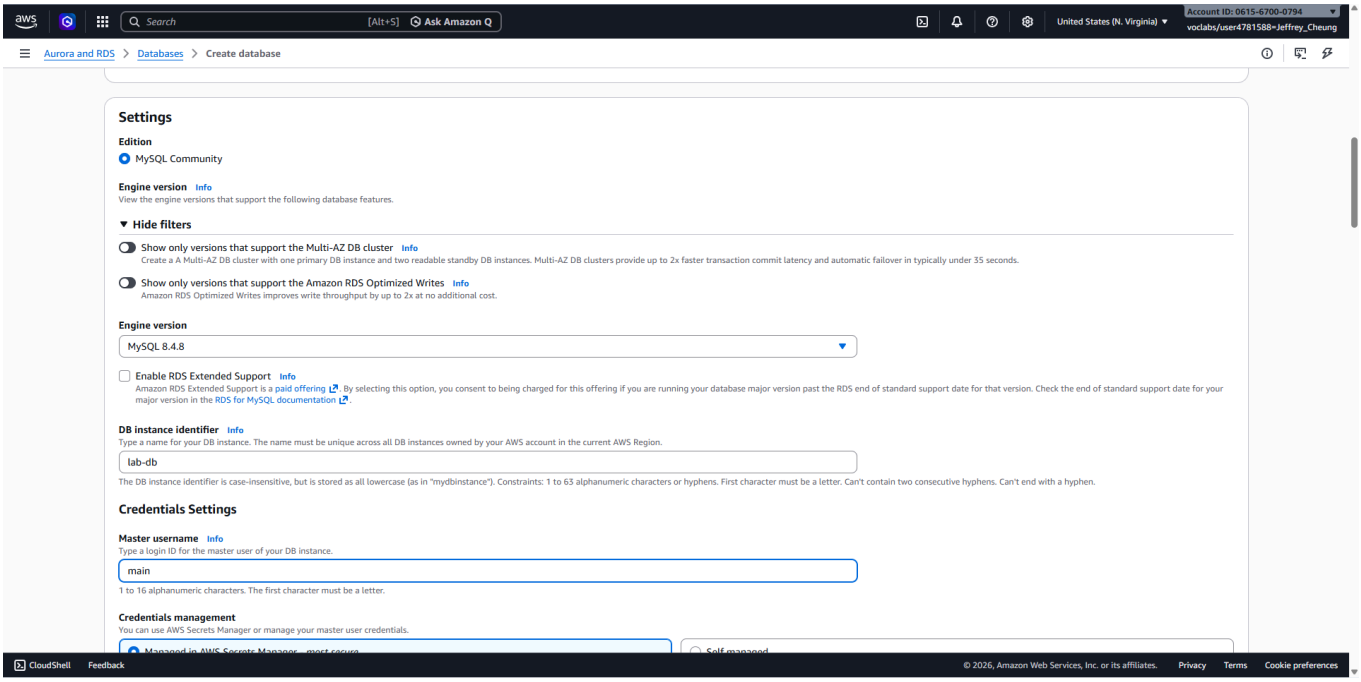


Figure 06. Confirm password: lab-password

Instance Configuration

Under DB instance class:

Select Burstable classes

Choose db.t3.micro

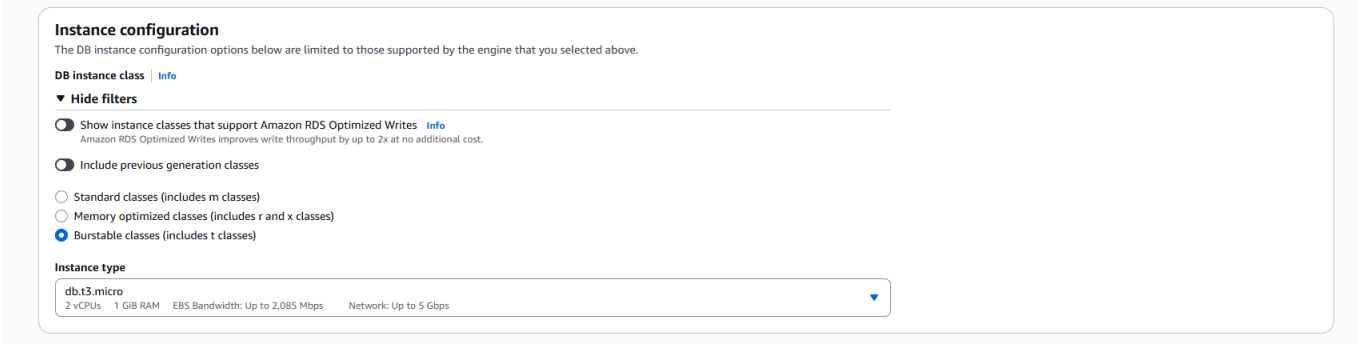


Figure 07. Choose db.t3.micro

Storage Configuration

Configure:

Storage type: General Purpose (SSD)

Allocated storage: 20 GiB

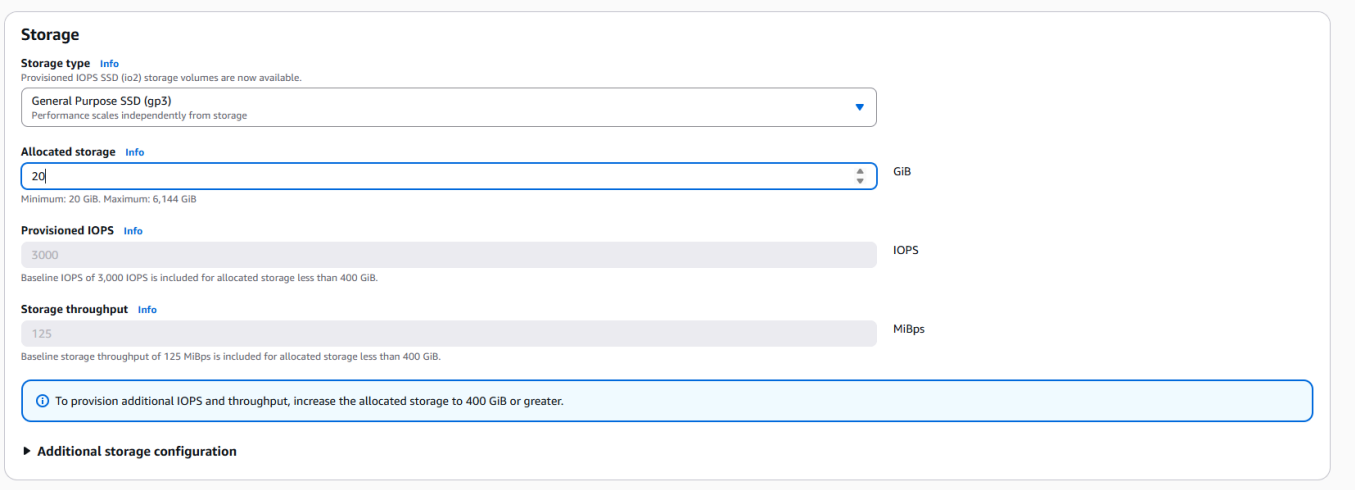


Figure 08. Allocated storage: 20 GiB

Connectivity

Configure:

VPC: Lab VPC

Security group: DB Security Group

Remove the default security group if present

Additional Settings

Expand Monitoring and disable:

Enable Enhanced Monitoring

Under Additional configuration, configure:

Initial database name: lab

Disable Automatic backups

Disable Encryption

Choose Create database.

Wait for Deployment

Select the lab-db database.

Wait until the database status changes to Modifying or Available.

Deployment may take several minutes because a Multi-AZ configuration is being created.

Open the Connectivity & Security section.

Copy the database Endpoint and save it for later use.

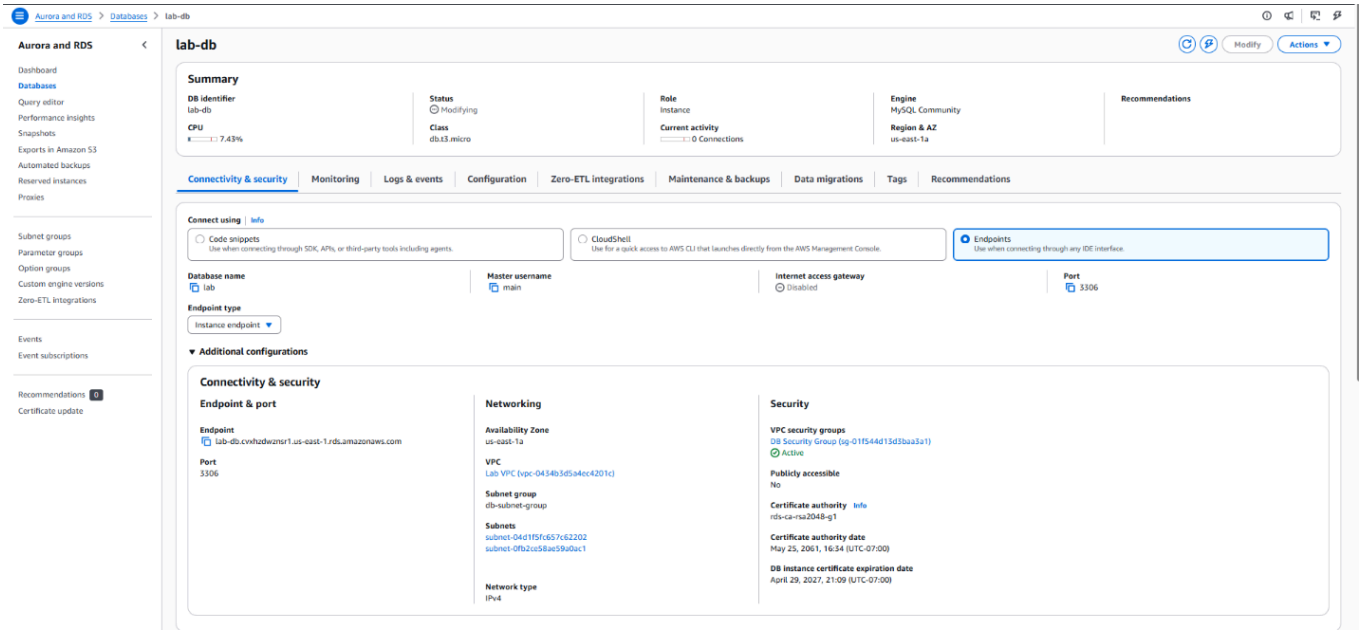


Figure 09. Copy the database Endpoint and save it for later use.

Example: lab-db.xxxx.us-east-1.rds.amazonaws.com

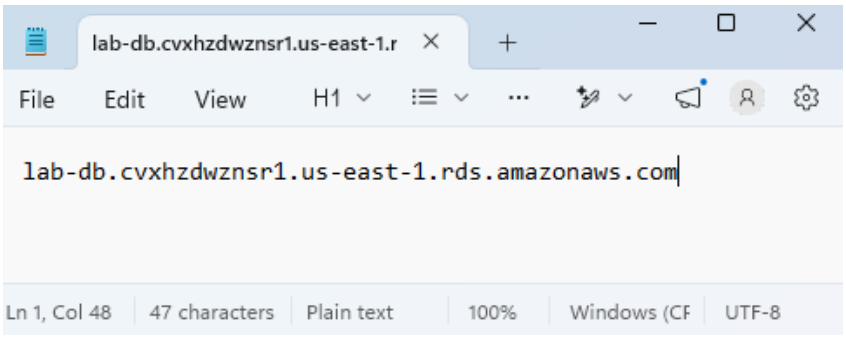


Figure 10. Example: lab-db.xxxx.us-east-1.rds.amazonaws.com

Task 4: Connect the Application to the Database

Obtain the WebServer IP address from the AWS Details panel.

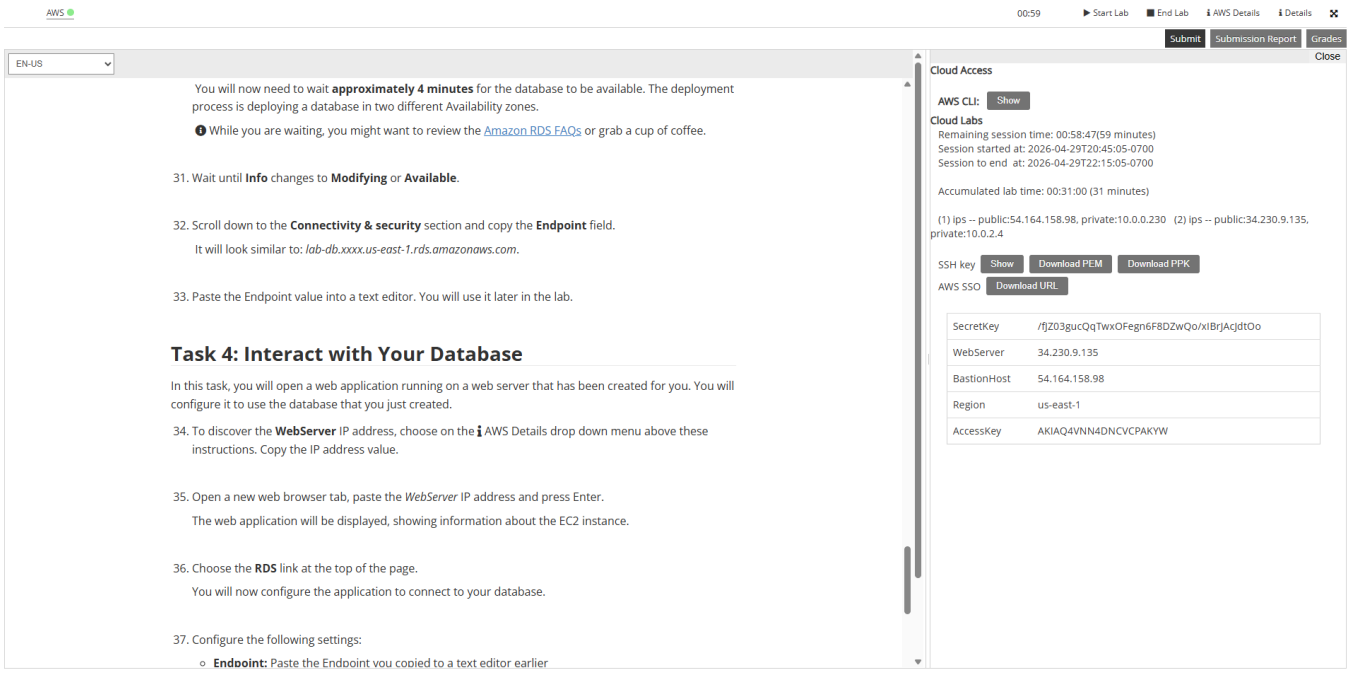


Figure 11. Obtain the WebServer IP address from the AWS Details panel.

Open a new browser tab and navigate to the WebServer IP address.

Verify that the web application loads successfully.

Select the RDS link at the top of the page.

Configure Database Access

Enter the following information:

Endpoint: Database endpoint copied earlier

Database: lab

Username: main

Password: lab-password

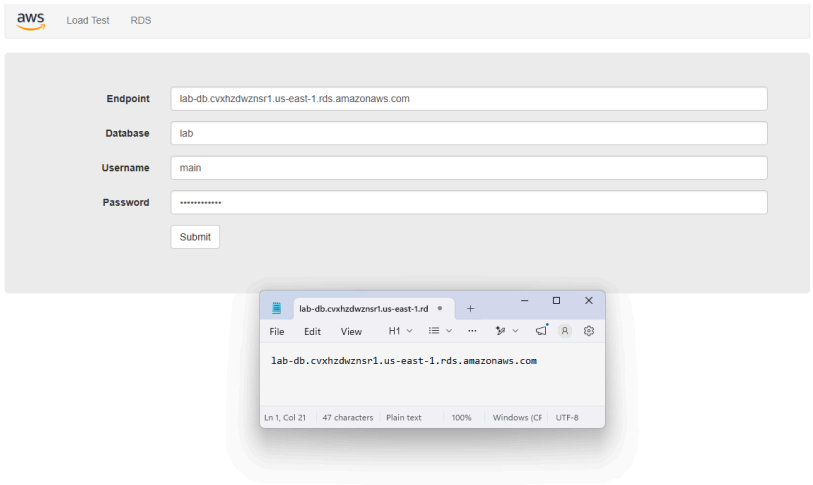


Figure 12. Password: lab-password

Choose Submit.

Wait for the application to initialize the database connection.

Once complete, the Address Book application will appear.

Test Database Functionality

Create a new contact.

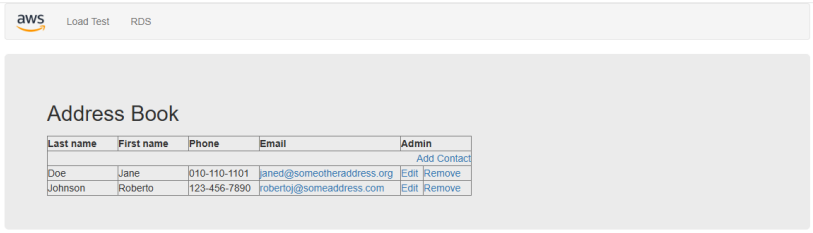


Figure 13. Create a new contact.

Edit an existing contact.

Delete a contact.

Verify that all changes are saved successfully.

The application is now using the Amazon RDS database to store data, and the database is automatically replicated across multiple Availability Zones for improved availability and durability.

Lab 6: Scale and Load balance Your Architecture

Task 1: Create an AMI for Auto Scaling

Open EC2 from the AWS Management Console.

Select Instances and verify that Web Server 1 shows 2/2 Status Checks Passed.

Select Web Server 1.

Choose:

Actions → Image and templates → Create image

Configure:

Image name: WebServerAMI

Image description: Lab AMI for Web Server

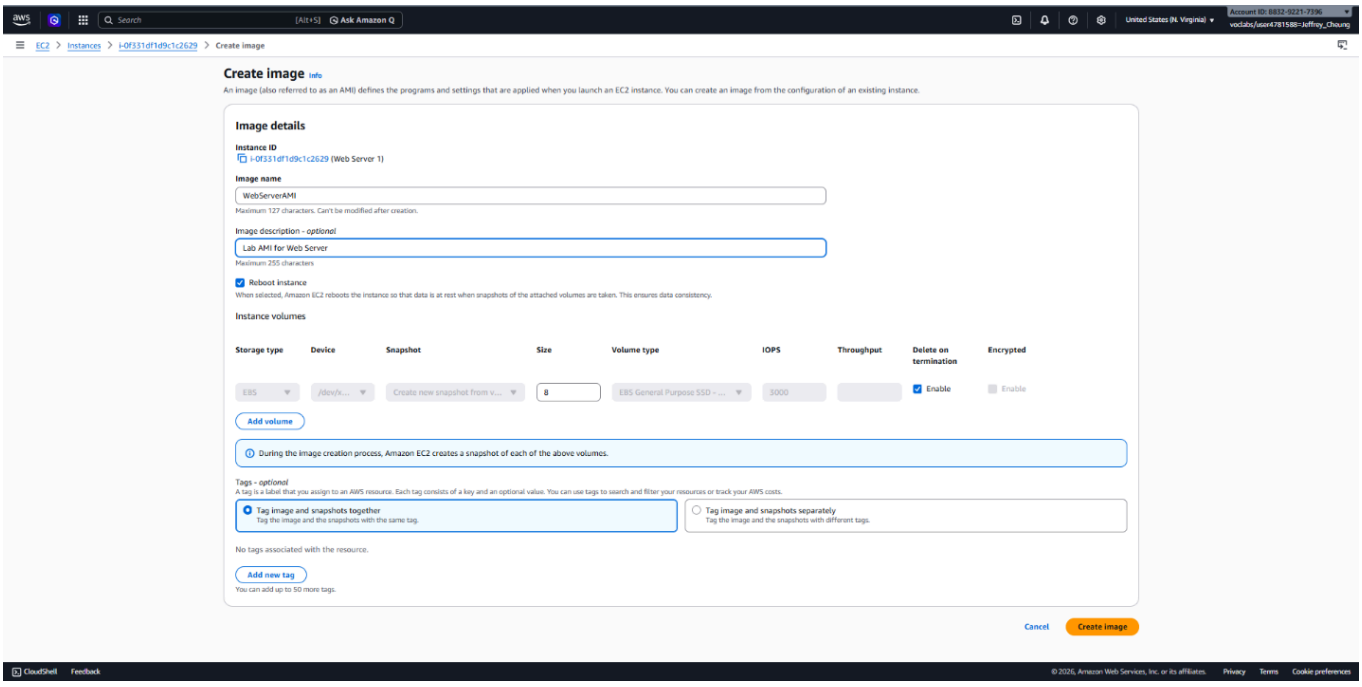


Figure 14. Image description: Lab AMI for Web Server

Choose Create image.

Record the AMI ID displayed in the confirmation message.

This AMI captures the current state of the web server and will be used to launch identical instances through Auto Scaling.

Task 2: Create a Target Group and Load Balancer

Create a Target Group

In the EC2 console, select Target Groups.

Choose Create target group.

Configure:

Target type: Instances

Target group name: LabGroup

VPC: Lab VPC

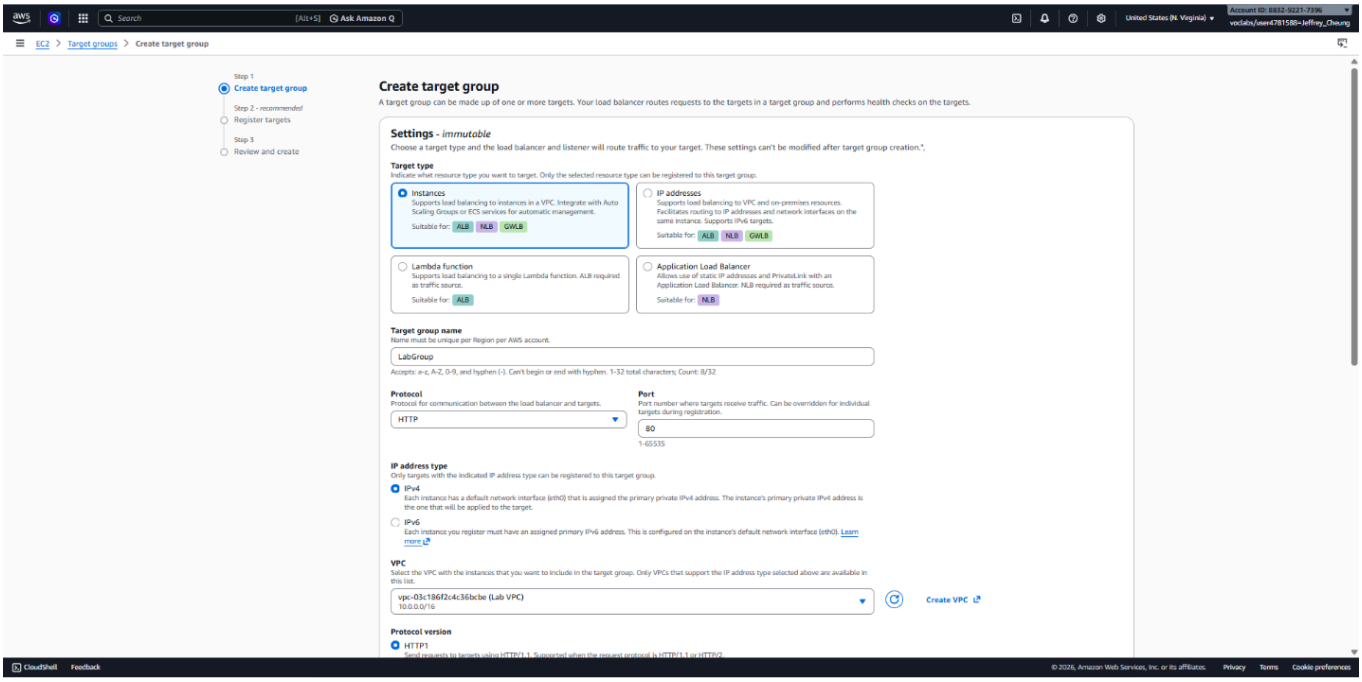


Figure 15. VPC: Lab VPC

Choose Next.

Skip target registration since no Auto Scaling instances exist yet.

Choose Create target group.

Create an Application Load Balancer

Select Load Balancers.

Choose Create load balancer.

Under Application Load Balancer, select Create.

Configure:

Load balancer name: LabELB

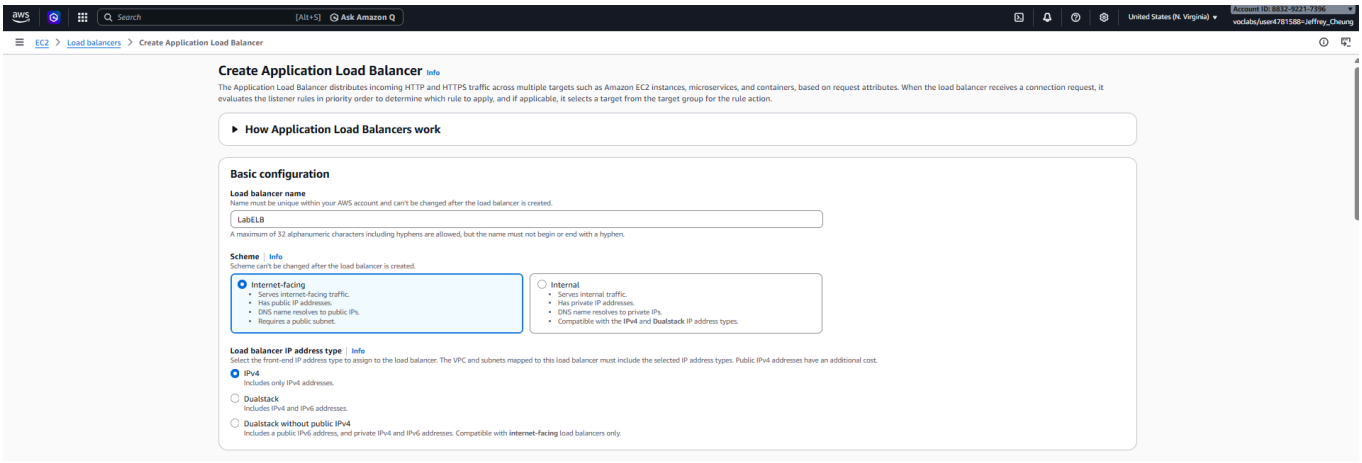


Figure 16. Load balancer name: LabELB

Under Network mapping:

VPC: Lab VPC

Select Public Subnet 1 in the first Availability Zone

Select Public Subnet 2 in the second Availability Zone

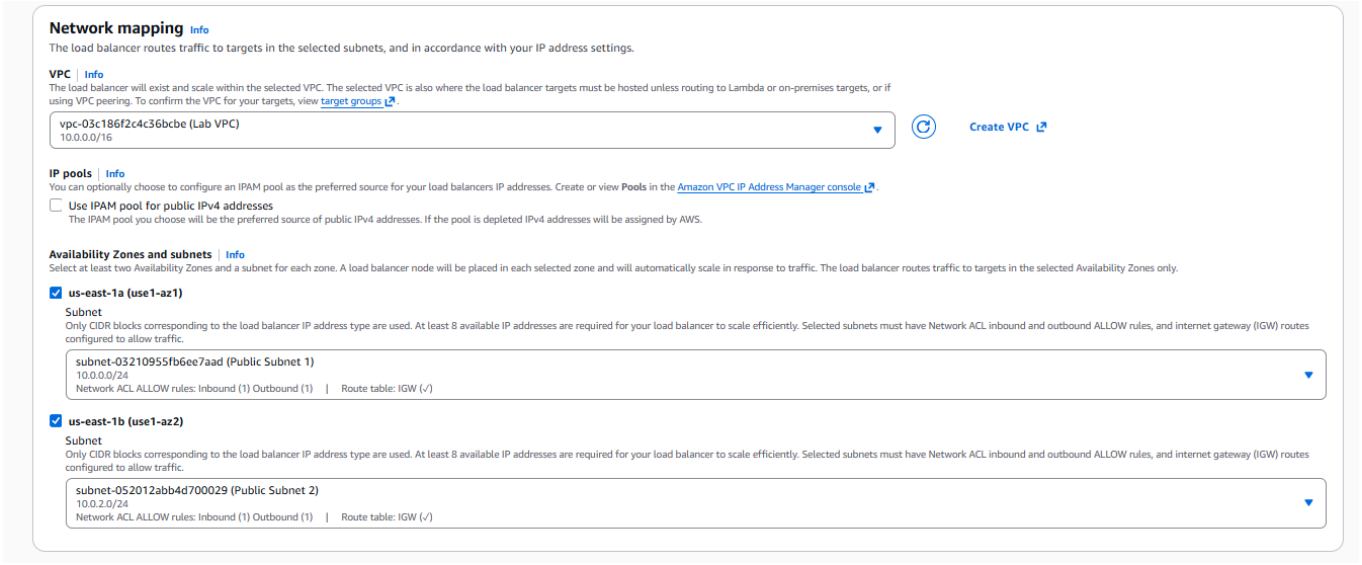


Figure 17. Select Public Subnet 2 in the second Availability Zone

Under Security Groups:

Select Web Security Group

Remove the default security group

Configure the HTTP listener:

Default action: Forward to LabGroup

Choose Create load balancer.

Once creation is complete, select View load balancer.

The load balancer may still show a status of Provisioning, which is expected.

Task 3: Create a Launch Template and Auto Scaling Group

Create a Launch Template

Select Launch Templates.

Choose Create launch template.

Configure:

Launch template name: LabConfig

Enable Auto Scaling guidance

Under My AMIs, select:

AMI: WebServerAMI

Configure:

Instance type: t2.micro

Key pair: vockey

Under Security Groups:

Select Existing security group

Choose Web Security Group

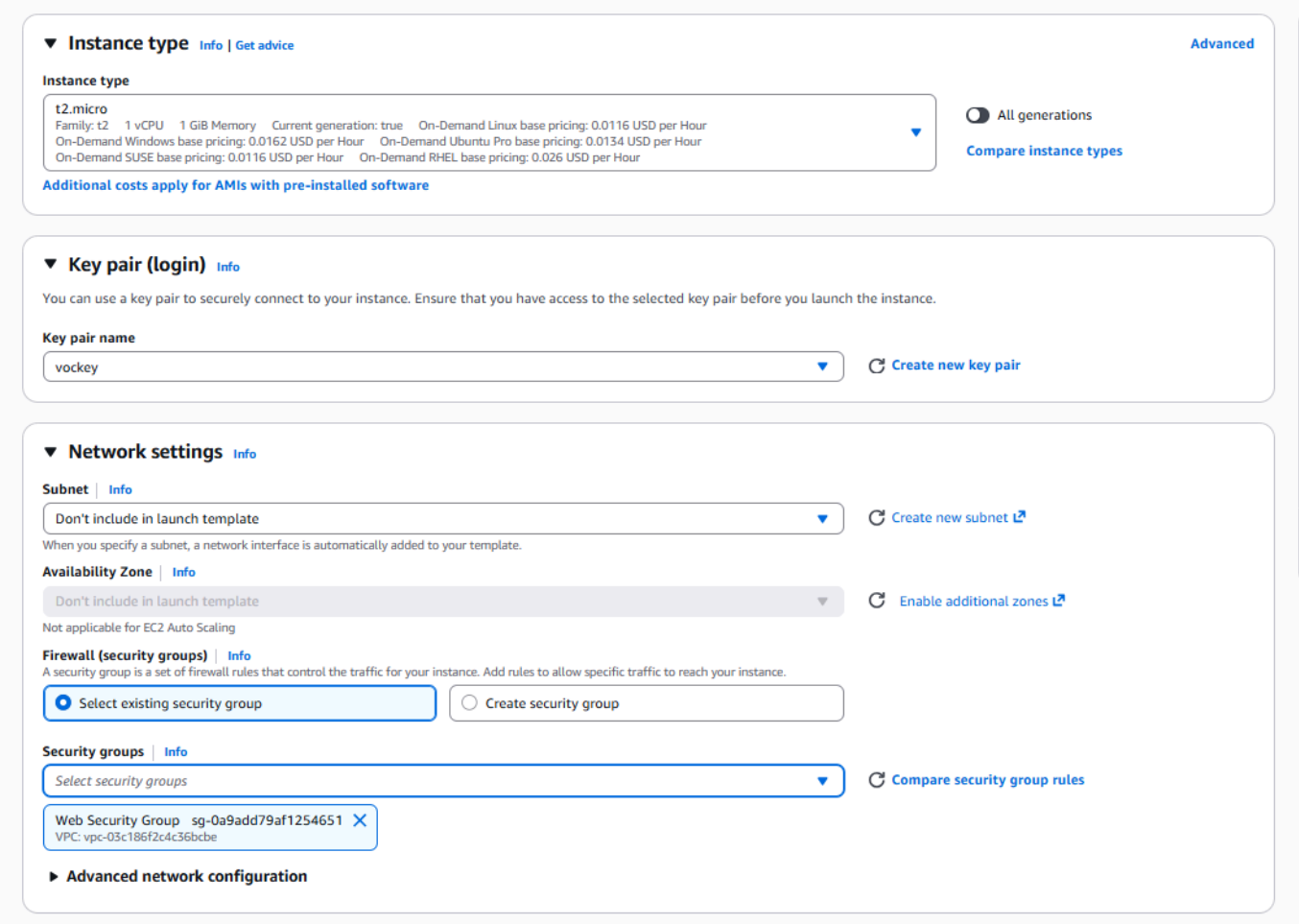


Figure 18. Choose Web Security Group

Expand Advanced Details.

Enable Detailed CloudWatch Monitoring.

Choose Create launch template.

Create the Auto Scaling Group

Open the newly created LabConfig template.

Choose:

Actions → Create Auto Scaling Group

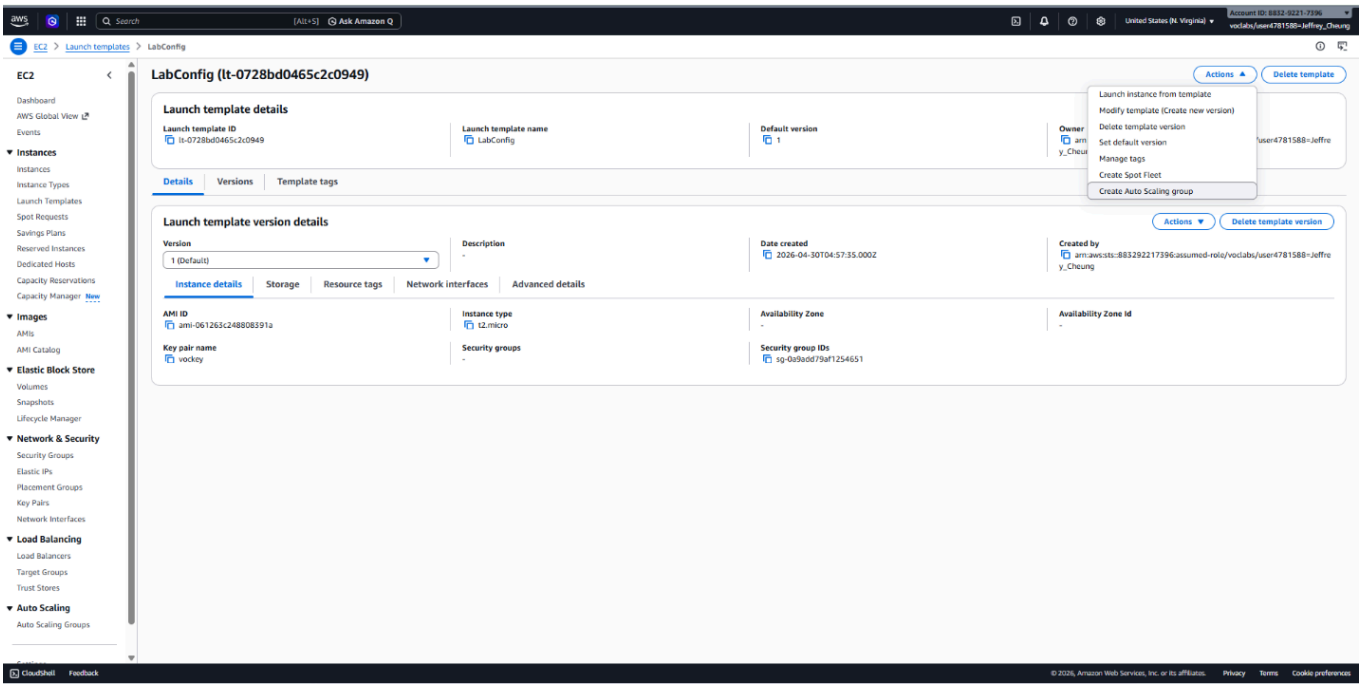


Figure 19. Actions → Create Auto Scaling Group

Step 1: Basic Configuration

Configure:

Auto Scaling group name: Lab Auto Scaling Group

Launch template: LabConfig

Choose Next.

Step 2: Networking

Configure:

VPC: Lab VPC

Subnets: Private Subnet 1 and Private Subnet 2

Choose Next.

Step 3: Load Balancing

Select:

Attach to an existing load balancer

Target group: LabGroup

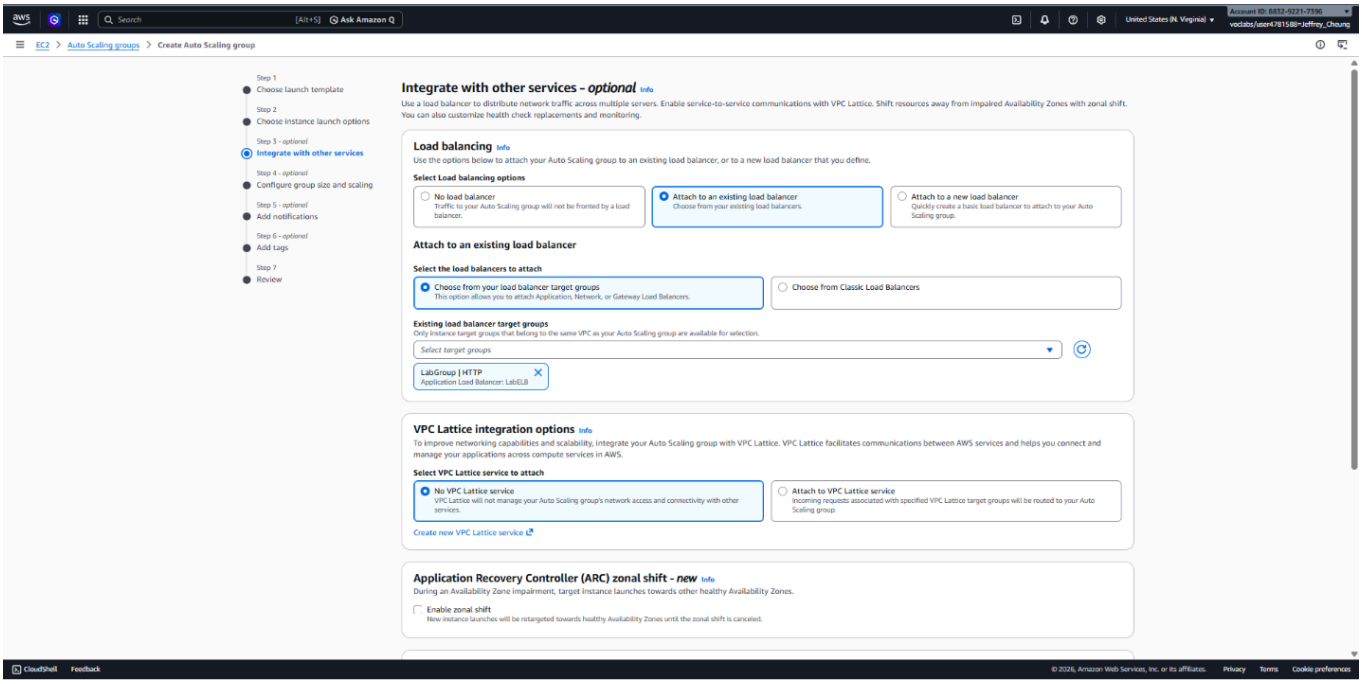


Figure 20. Target group: LabGroup

Enable:

Group metrics collection within CloudWatch

Choose Next.

Step 4: Scaling Configuration

Configure:

Desired capacity: 2

Minimum capacity: 2

Maximum capacity: 6

Configure the scaling policy:

Policy type: Target Tracking

Policy name: LabScalingPolicy

Metric: Average CPU Utilization

Target value: 60%

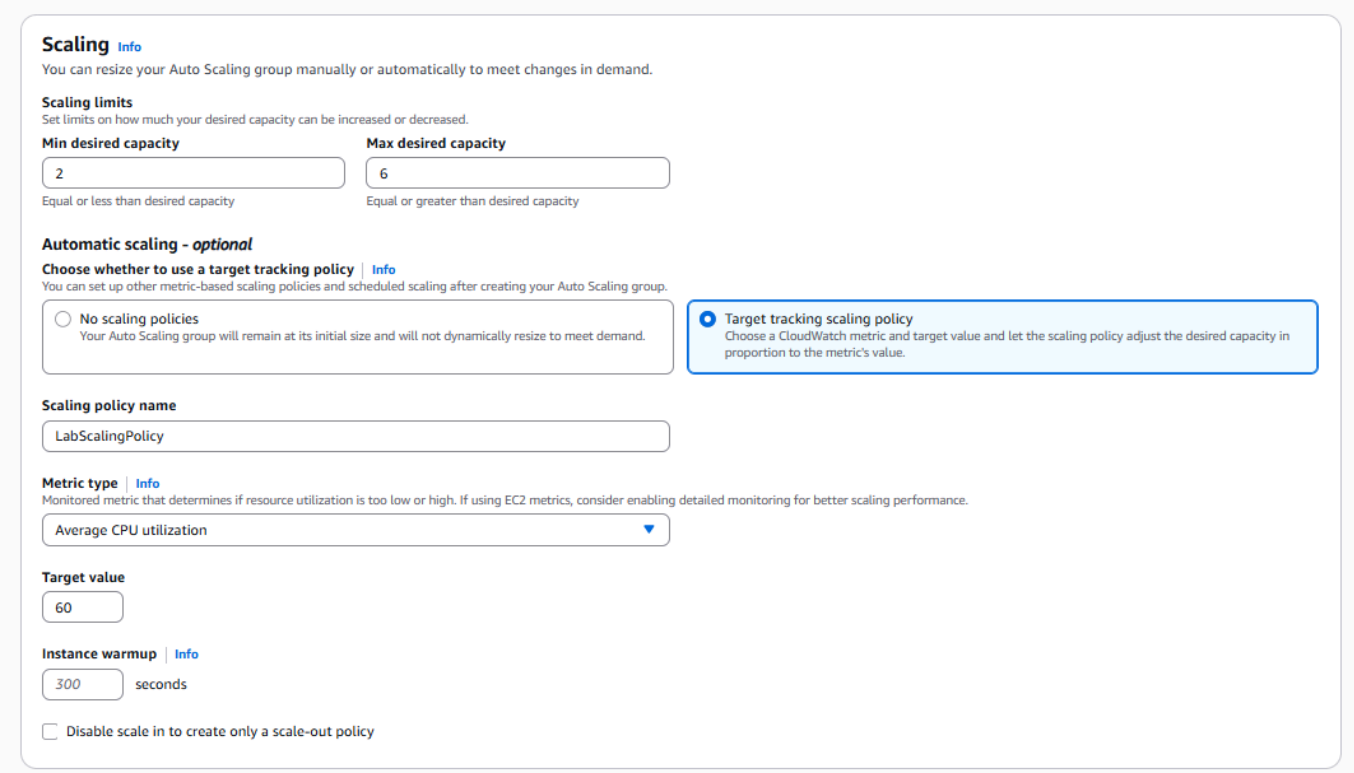


Figure 21. Target value: 60%

Choose Next.

Step 5: Notifications

Leave the default settings and choose Next.

Step 6: Tags

Add a tag:

Key: Name

Value: Lab Instance

Choose Next.

Review the configuration and choose Create Auto Scaling Group.

The group will automatically launch two instances to meet the desired capacity.

Task 4: Verify Load Balancing

Open Instances.

Confirm that two instances named Lab Instance have been launched.

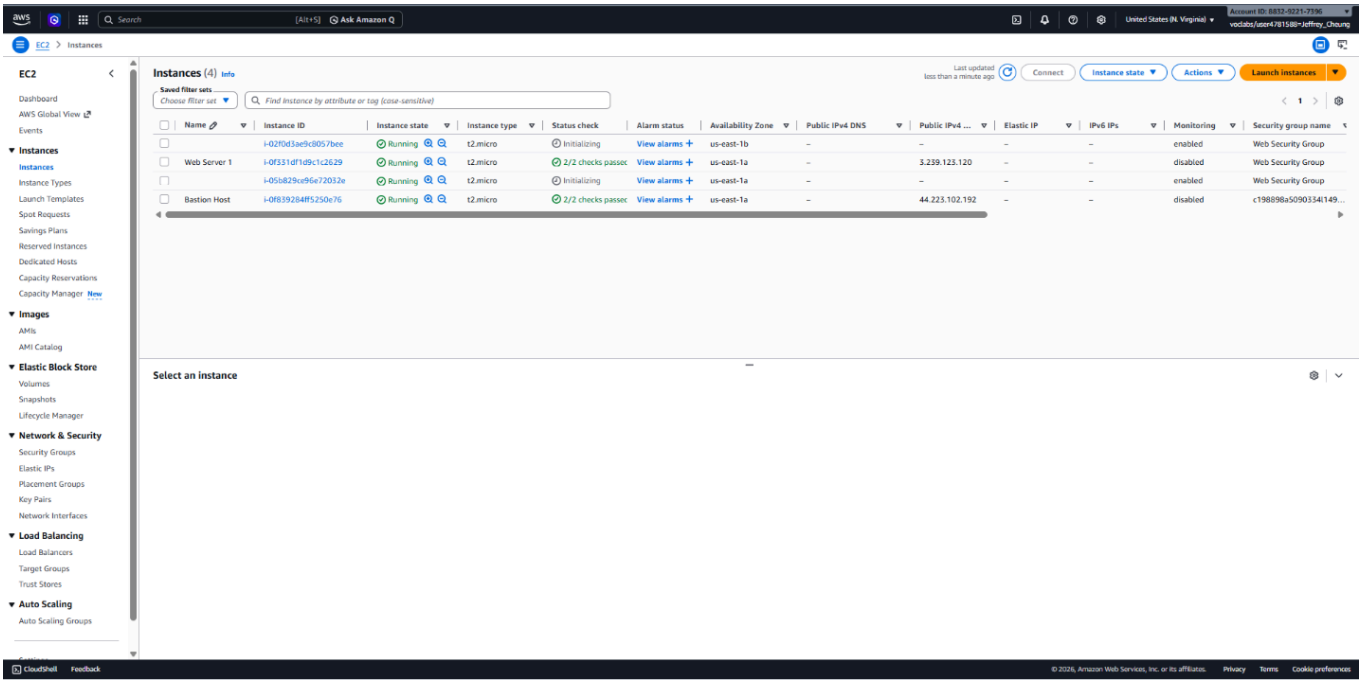


Figure 22. Confirm that two instances named Lab Instance have been launched.

Select Target Groups and open LabGroup.

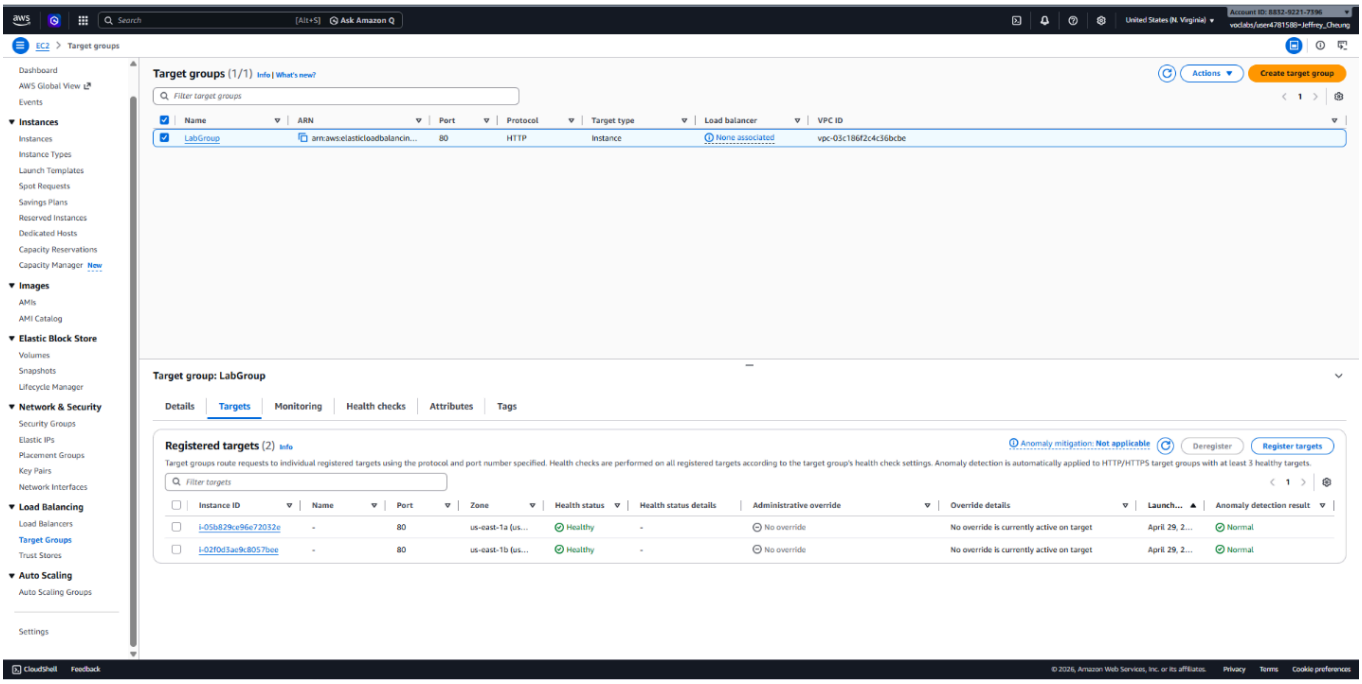


Figure 23. Select Target Groups and open LabGroup.

Open the Targets tab.

Wait until both instances display a status of Healthy.

Healthy targets have passed the load balancer health checks and are eligible to receive traffic.

Select Load Balancers.

Open LabELB.

Copy the load balancer DNS Name.

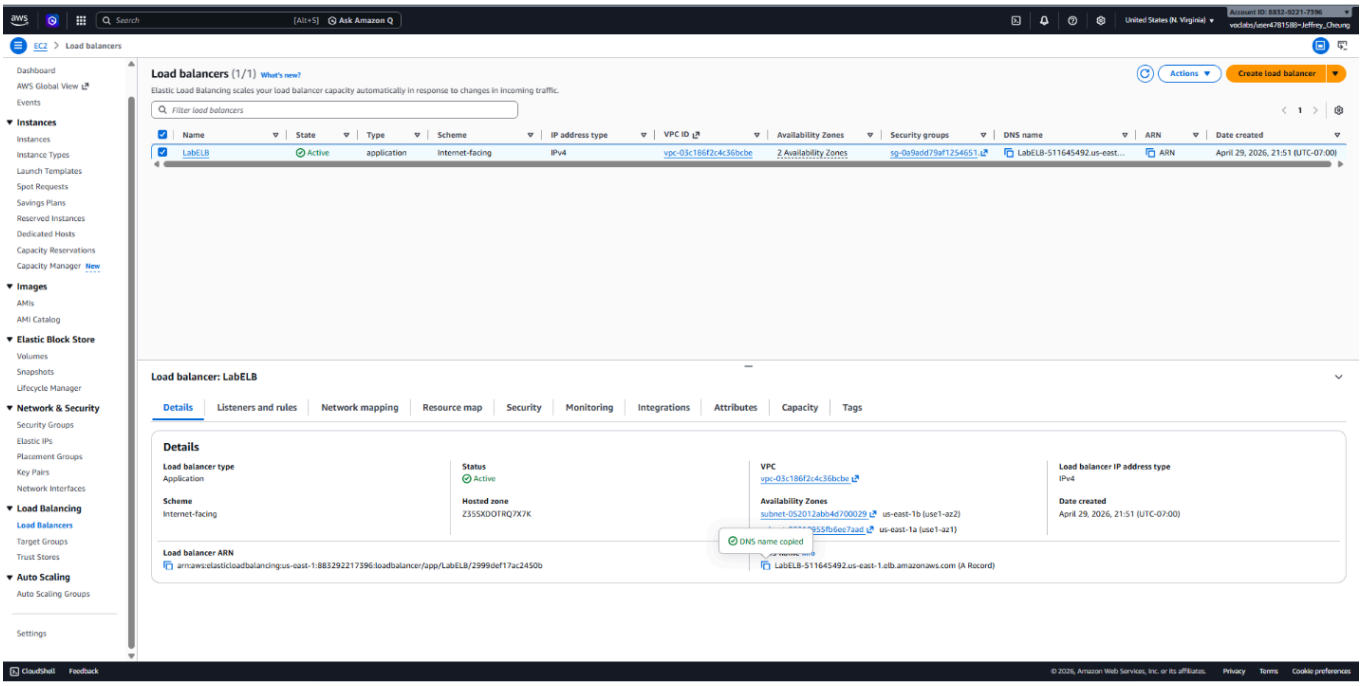


Figure 24. Copy the load balancer DNS Name.

Paste the DNS name into a web browser.

Verify that the application loads successfully.

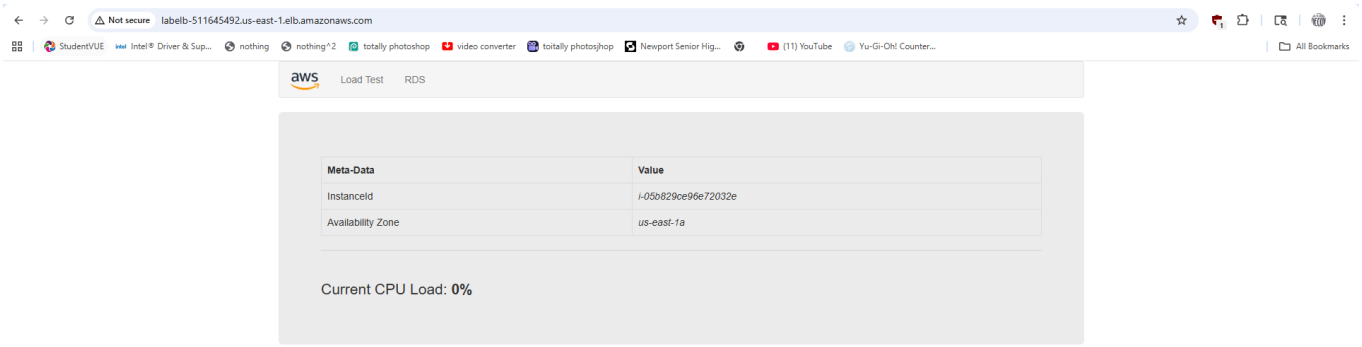


Figure 25. Verify that the application loads successfully.

This confirms that traffic is being routed through the load balancer to the Auto Scaling instances.

Task 5: Test Auto Scaling

Review Scaling Alarms

Open CloudWatch.

Select All Alarms.

Locate the two alarms automatically created by the Auto Scaling policy.

Open the alarm containing AlarmHigh in its name.

Verify that the alarm is currently in the OK state.

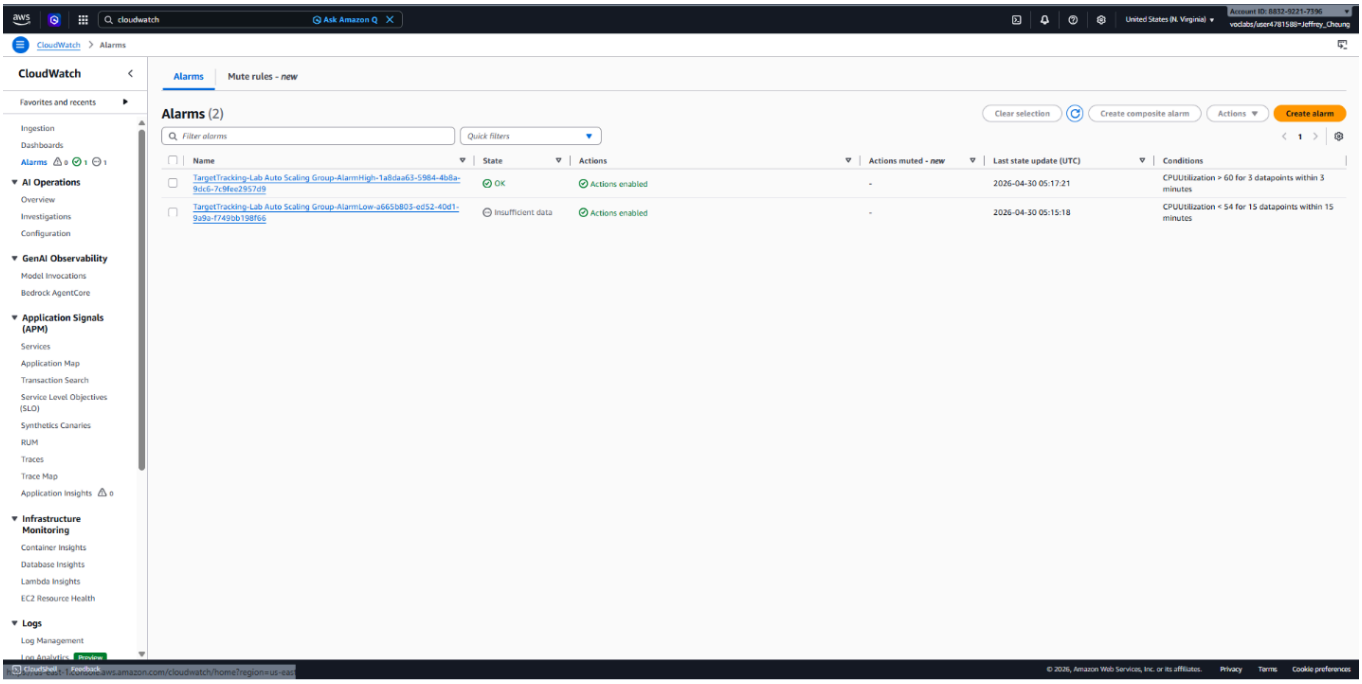


Figure 26. Verify that the alarm is currently in the OK state.

Generate Load

Return to the web application.

Select Load Test.

Leave the page open so the load continues to run.

Monitor Scaling Activity

Return to CloudWatch.

Refresh the alarms periodically.

Wait for:

AlarmHigh to change to In Alarm

CPU utilization to exceed the target threshold

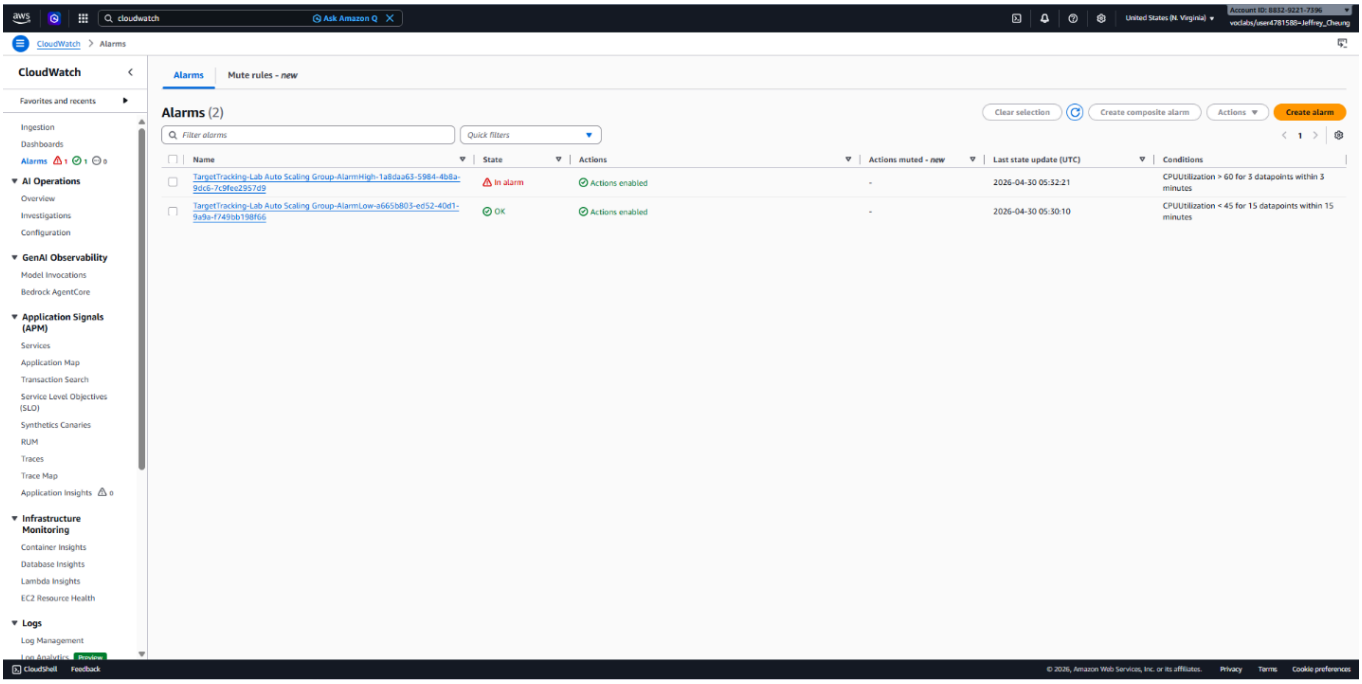


Figure 27. CPU utilization to exceed the target threshold

Once the alarm is triggered, Auto Scaling launches additional instances automatically.

Verify New Instances

Return to EC2 → Instances.

Confirm that more than two Lab Instance servers are running.

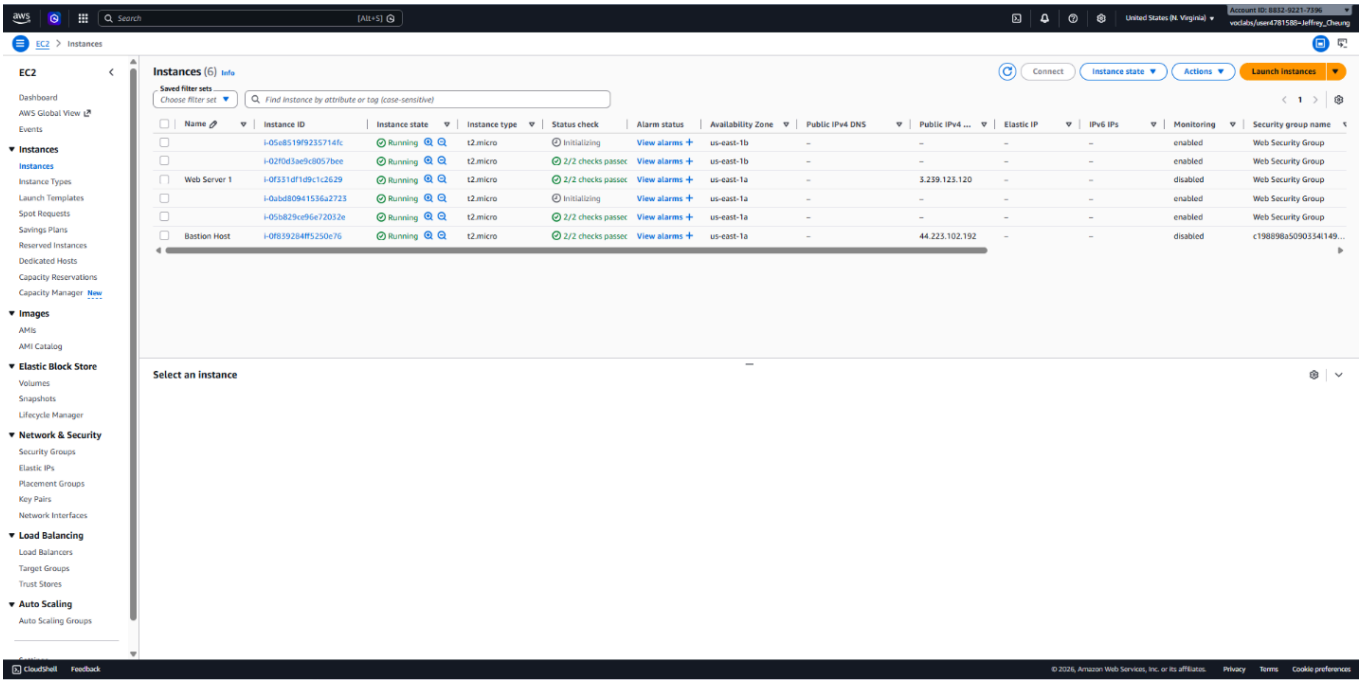


Figure 28. Confirm that more than two Lab Instance servers are running.

This demonstrates Auto Scaling responding automatically to increased demand.

Task 6: Terminate the Original Web Server

Return to Instances.

Select Web Server 1 only.

Choose:

Instance State → Terminate Instance

Confirm by selecting Terminate.

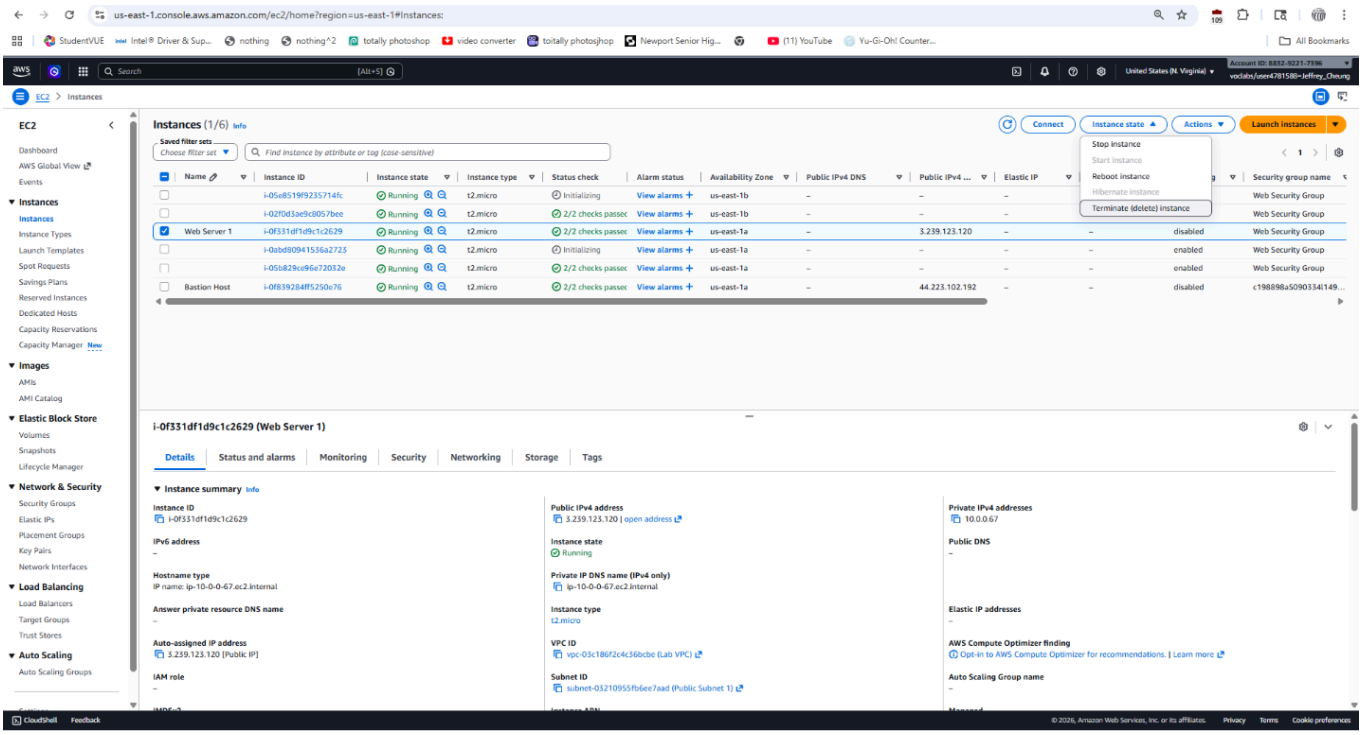


Figure 29. Confirm by selecting Terminate.

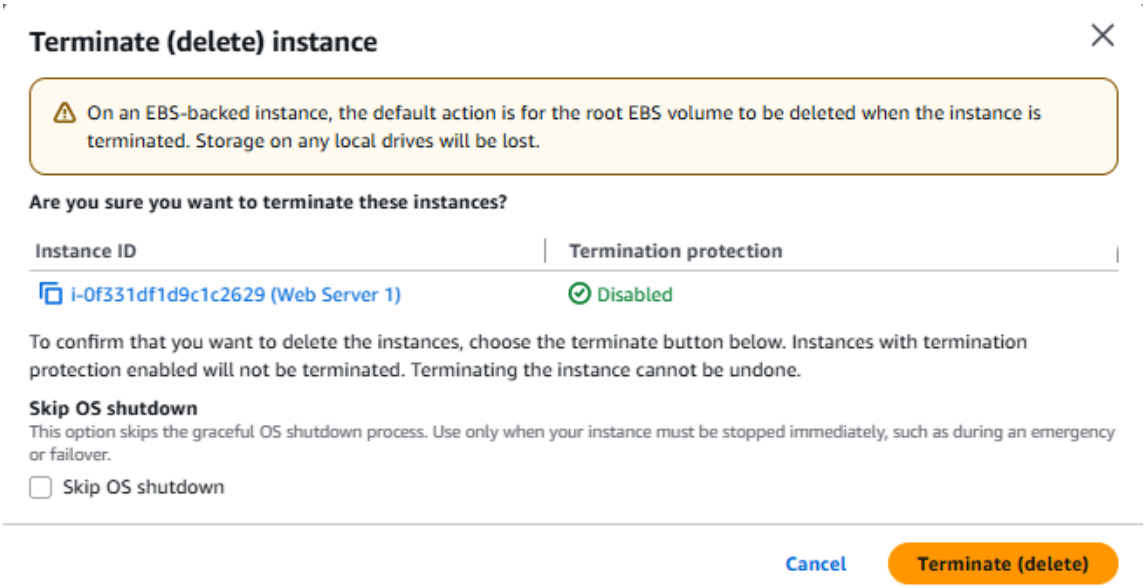


Figure 30. Confirm by selecting Terminate.

The original instance is no longer needed because its configuration has been captured in the AMI and is now managed through the Auto Scaling group.

Troubleshooting and Lessons Learned

The AWS Academy grading interface occasionally failed to register completed steps. Reviewing the grading status throughout the exercise reduced the need to retrace earlier work and helped identify steps that required repetition.

Conclusion

The exercises linked EBS storage, managed database connectivity, load balancing, and automated scaling. They provided practical experience with persistent data, application health checks, launch templates, and capacity adjustments driven by monitored demand.